

MSBA Job Matching

Grounded NLP decision support for career advisors

Real postings -> retrieved evidence -> role-fit triage -> human review

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The advisor must separate fit, uncertainty, and risk

A useful system improves a specific decision: forward, hold, or investigate.

FORWARD

Strong entry-level analytics fit supported by title, skills, and location evidence.

HOLD

Senior, engineering-heavy, or clearly misaligned roles require a conservative screen.

INVESTIGATE

Incomplete fields or missing authorization language require human follow-up.

Role fit and work authorization are related in workflow, but they are not the same prediction task.

The data is large, real, and reproducibly sampled

Transparent weak labels make experimentation feasible, but not equivalent to human ground truth.

785,741

original public job
records

100,000

stratified project
sample

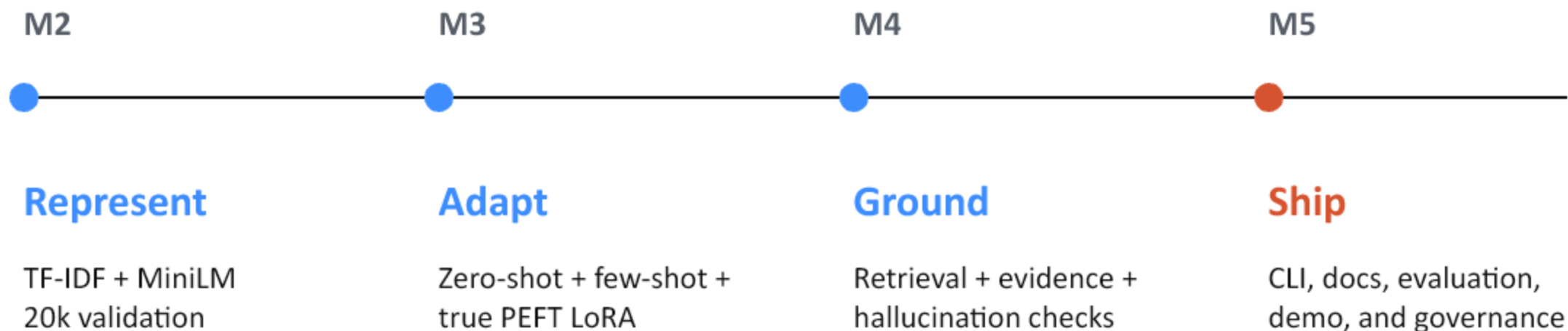
**80k /
20k**

training and held-out
validation

4

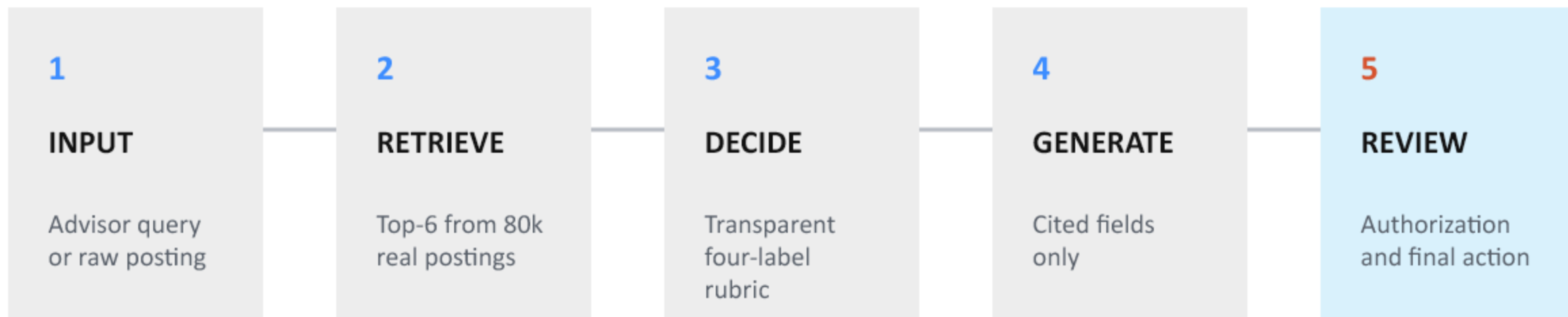
balanced role-fit
labels

Each milestone added a measured capability



Retrieval makes the final recommendation inspectable

Generation is constrained to source fields; authorization remains an evidence-gated branch.



Every posting-specific fact in the response can be traced to a retrieved record.

A normal run returns real postings, not just a label

Query prediction: high_fit | Action: advisor review before forwarding

INPUT Find entry-level US data analyst, business analyst, or BI roles for an MSBA student with SQL, Python, Excel, Tableau, and Power BI. Remote or Bay Area preferred.

RANK	POSTING	COMPANY / LOCATION	FIT	SCORE
1	Data Analyst	HireMatch Anywhere	high_fit	8.054
2	Data Analyst II, Bay Area - Remote	Astreya Anywhere	high_fit	7.951
3	Data Analyst (US REMOTE)	LeanTaaS Anywhere	high_fit	6.858

Review 6 retrieved postings first; they have analytics role-fit signals.

Authorization: verify separately unless explicit source text is present.

Edge cases change the action, not the safety boundary

ENTRY LEVEL ANALYTICS

high_fit

Forward candidates after advisor review

SENIOR ENGINEERING FILTER

low_fit

Hold a role that is too senior or engineering-heavy

AUTHORIZATION MISSING

unclear

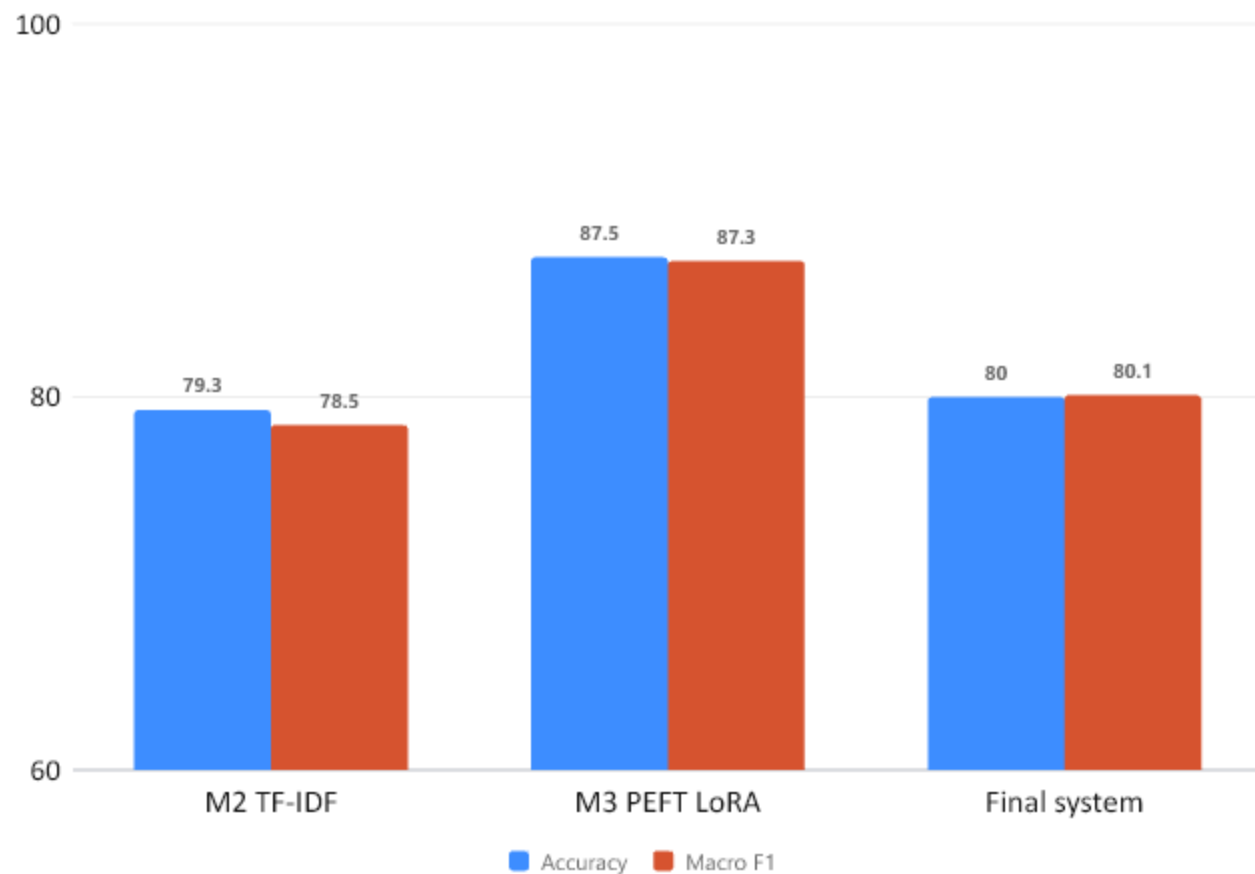
Explicit negative authorization text surfaced

THIN METADATA

unclear

Investigate an incomplete posting

The final system preserves quality and adds evidence



80.1% final macro F1

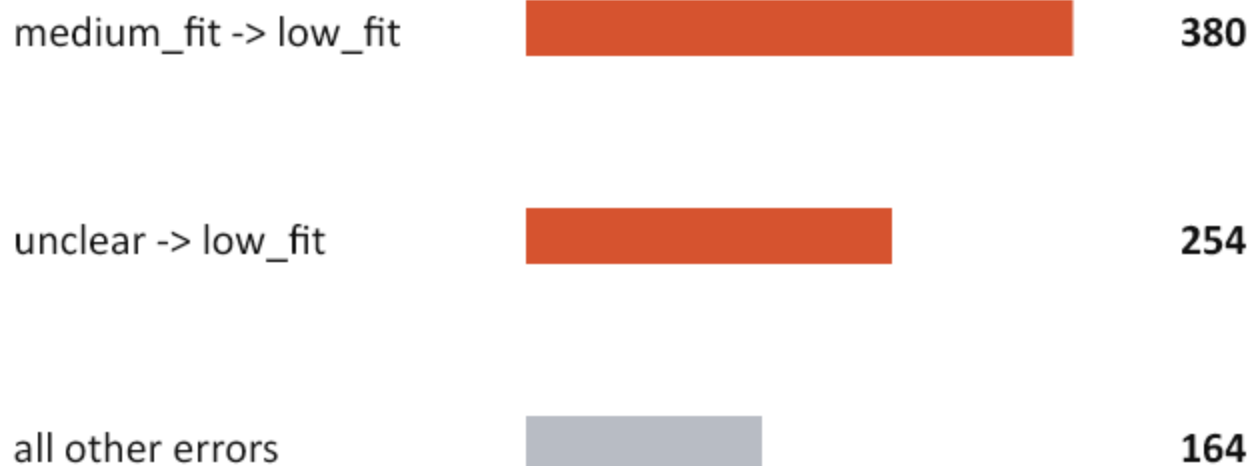
90.0% support hit@5

0 unsupported
authorization claims

LoRA wins classification; the final system wins the workflow tradeoff.

Errors require a conservative deployment policy

WHERE ERRORS CONCENTRATE



DEPLOYMENT RULE

**Prioritize high-fit candidates.
Route medium, unclear, and
authorization-sensitive cases to
human review.**

No automatic rejection
No unsupported authorization claim
No private student data in the prototype

Pilot as decision support, then earn the right to automate more

Next measures

High-fit precision
Advisor time saved
Override rate
Authorization extraction accuracy

Use de-identified Career Center postings and human-reviewed labels.